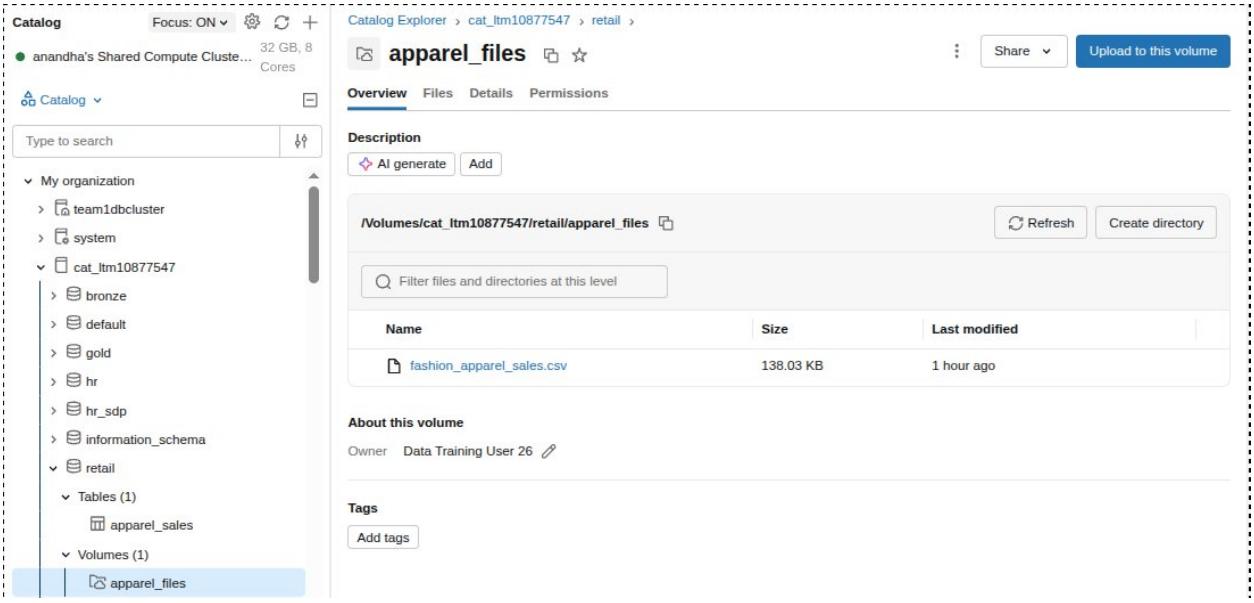
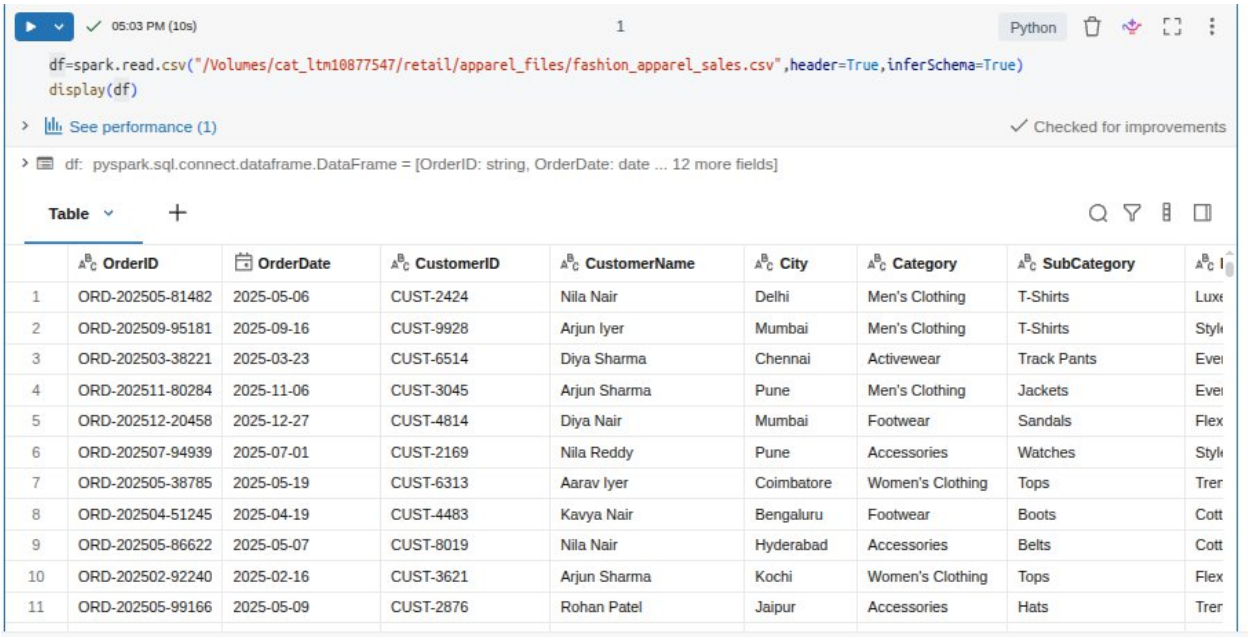
Soumiya Ayyappan_W10_D2

Step 1:



Step 2:



05:03 PM (10s) 1 Python

```
df=spark.read.csv("/Volumes/cat_ltm10877547/retail/apparel_files/fashion_apparel_sales.csv",header=True,inferSchema=True)
display(df)
```

See performance (1) Checked for improvements

df: pyspark.sql.connect.dataframe.DataFrame = [OrderID: string, OrderDate: date ... 12 more fields]

Table

	OrderID	OrderDate	CustomerID	CustomerName	City	Category	SubCategory	
1	ORD-202505-81482	2025-05-06	CUST-2424	Nila Nair	Delhi	Men's Clothing	T-Shirts	Luxe
2	ORD-202509-95181	2025-09-16	CUST-9928	Arjun Iyer	Mumbai	Men's Clothing	T-Shirts	Styl
3	ORD-202503-38221	2025-03-23	CUST-6514	Diya Sharma	Chennai	Activewear	Track Pants	Ever
4	ORD-202511-80284	2025-11-06	CUST-3045	Arjun Sharma	Pune	Men's Clothing	Jackets	Ever
5	ORD-202512-20458	2025-12-27	CUST-4814	Diya Nair	Mumbai	Footwear	Sandals	Flex
6	ORD-202507-94939	2025-07-01	CUST-2169	Nila Reddy	Pune	Accessories	Watches	Styl
7	ORD-202505-38785	2025-05-19	CUST-6313	Aarav Iyer	Coimbatore	Women's Clothing	Tops	Trer
8	ORD-202504-51245	2025-04-19	CUST-4483	Kavya Nair	Bengaluru	Footwear	Boots	Cott
9	ORD-202505-86622	2025-05-07	CUST-8019	Nila Nair	Hyderabad	Accessories	Belts	Cott
10	ORD-202502-92240	2025-02-16	CUST-3621	Arjun Sharma	Kochi	Women's Clothing	Tops	Flex
11	ORD-202505-99166	2025-05-09	CUST-2876	Rohan Patel	Jaipur	Accessories	Hats	Trer

05:04 PM (1s)

```
df.printSchema()
print(df.count())
```

See performance (1)

```
root
 |-- OrderID: string (nullable = true)
 |-- OrderDate: date (nullable = true)
 |-- CustomerID: string (nullable = true)
 |-- CustomerName: string (nullable = true)
 |-- City: string (nullable = true)
 |-- Category: string (nullable = true)
 |-- SubCategory: string (nullable = true)
 |-- Brand: string (nullable = true)
 |-- ProductID: string (nullable = true)
 |-- ProductName: string (nullable = true)
 |-- Quantity: integer (nullable = true)
 |-- UnitPrice: double (nullable = true)
 |-- DiscountRate: double (nullable = true)
 |-- SalesAmount: double (nullable = true)

1000
```

05:11 PM (3s) 3

```
df.write.format("delta").mode("overwrite").option("overwriteSchema","true").saveAsTable("cat_ltm10877547.retail.apparel_sales")
```

See performance (1) Checked for improvements

05:17 PM (1s) 4

```
%sql
select * from cat_ltm10877547.retail.apparel_sales limit 10;
```

See performance (1) Checked for improvements

Table

	OrderID	OrderDate	CustomerID	CustomerName	City	Category	SubCategory	
1	ORD-202505-81482	2025-05-06	CUST-2424	Nila Nair	Delhi	Men's Clothing	T-Shirts	Luxe
2	ORD-202509-95181	2025-09-16	CUST-9928	Arjun Iyer	Mumbai	Men's Clothing	T-Shirts	Styl
3	ORD-202503-38221	2025-03-23	CUST-6514	Diya Sharma	Chennai	Activewear	Track Pants	Ever
4	ORD-202511-80284	2025-11-06	CUST-3045	Arjun Sharma	Pune	Men's Clothing	Jackets	Ever
5	ORD-202512-20458	2025-12-27	CUST-4814	Diya Nair	Mumbai	Footwear	Sandals	Flex
6	ORD-202507-94939	2025-07-01	CUST-2169	Nila Reddy	Pune	Accessories	Watches	Styl
7	ORD-202505-38785	2025-05-19	CUST-6313	Aarav Iyer	Coimbatore	Women's Clothing	Tops	Trer

Step 3:

05:35 PM (1s) 5: **total sales**

```
%sql
select round(sum(SalesAmount),2) as total_sales from cat_ltm10877547.retail.apparel_sales;
```

See performance (1)

Table

	1.2 total_sales
1	8590530.74

05:43 PM (1s) 6: **Sales by Category**

```
%sql
select Category,round(sum(SalesAmount),2) as category_revenue from cat_ltm10877547.retail.apparel_sales group by Category order by
category_revenue desc limit 10;
```

See performance (1) Checked for improvements

Table

	Category	1.2 category_revenue
1	Accessories	1998776.45
2	Men's Clothing	1711919.26
3	Footwear	1695963.52
4	Women's Clothing	1659488.04
5	Activewear	1524383.47

05:48 PM (1s) 7: **Top brands** SQL

```
%sql
select brand, round(sum(salesamount),2) as category_revenue from cat_ltm10877547.retail.apparel_sales group by brand order by category_revenue
desc limit 10;
```

See performance (1) Checked for improvements

Table

	brand	1.2 category_revenue
1	LuxeLane	1352133.34
2	FlexWear	1346545.31
3	CottonCraft	1303496.8
4	StyleNova	1222636.93
5	UrbanThread	1178371.6
6	Everyday Co.	1128685
7	TrendHouse	1058661.76

05:50 PM (1s) 8: **Sales by city**

```
%sql
select city, round(sum(salesamount),2) as category_revenue from cat_ltm10877547.retail.apparel_sales group by city order by category_revenue
desc limit 10;
```

See performance (1) Checked for improvements

Table +

	city	category_revenue
1	Bengaluru	961218.14
2	Hyderabad	960781.79
3	Pune	948947.67
4	Coimbatore	928267.66
5	Ahmedabad	898304.38
6	Chennai	817849.17
7	Jaipur	780082.23
8	Kochi	779428.83
9	Delhi	771054.61
10	Mumbai	744596.26

05:57 PM (2s) 9: **Average Selling Price**

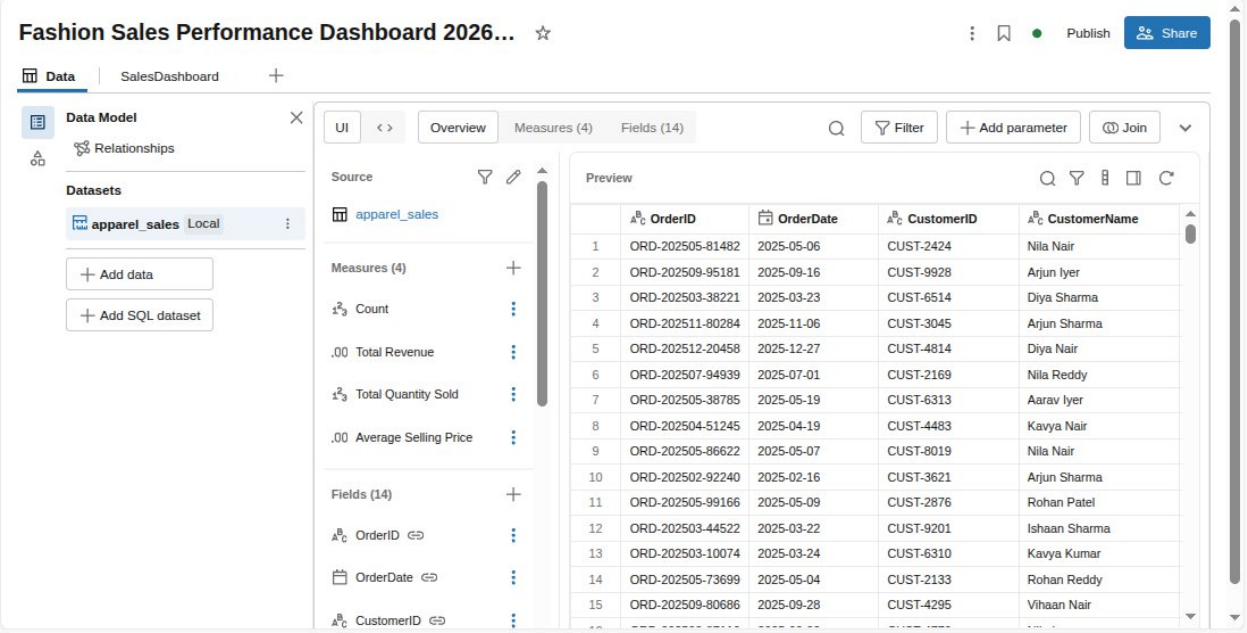
```
%sql
select round(avg(unitprice),2) as average_selling_price from cat_ltm10877547.retail.apparel_sales;
```

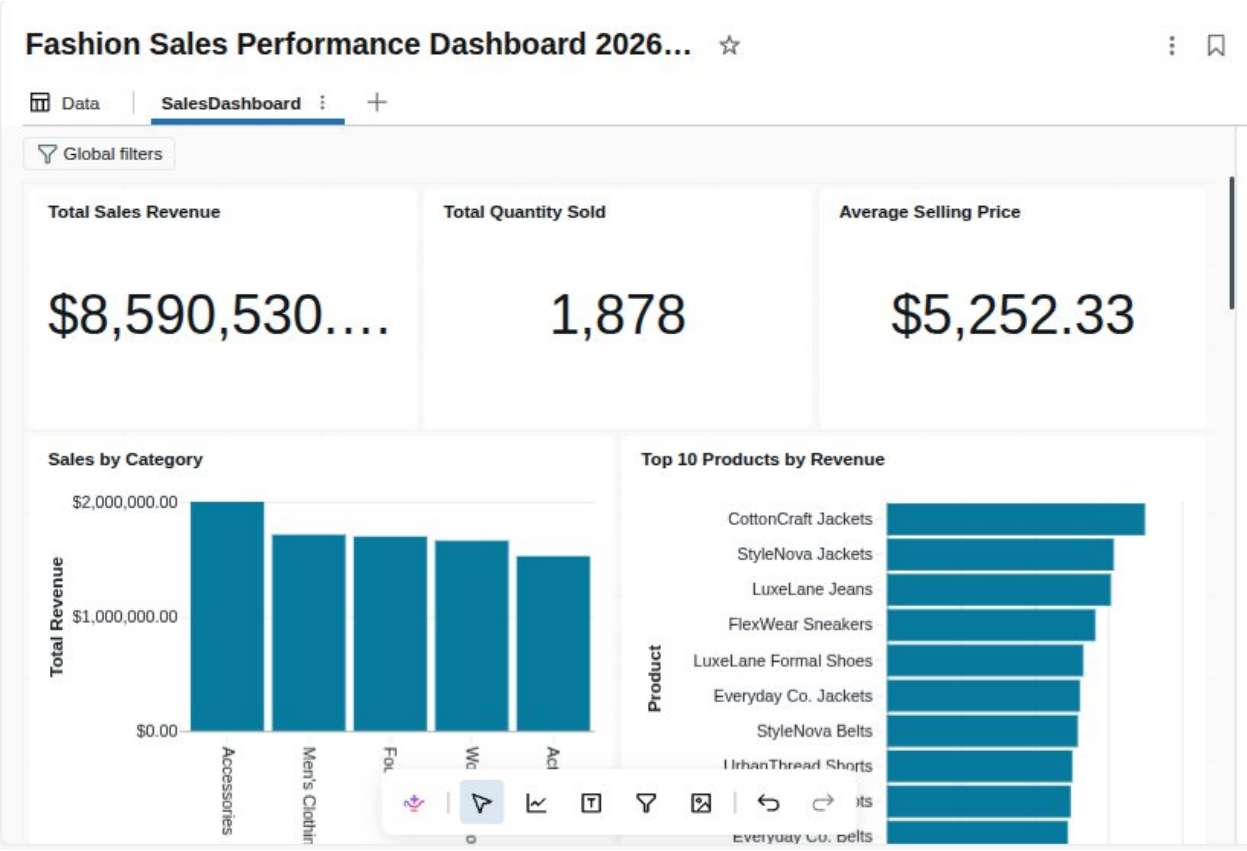
See performance (1)

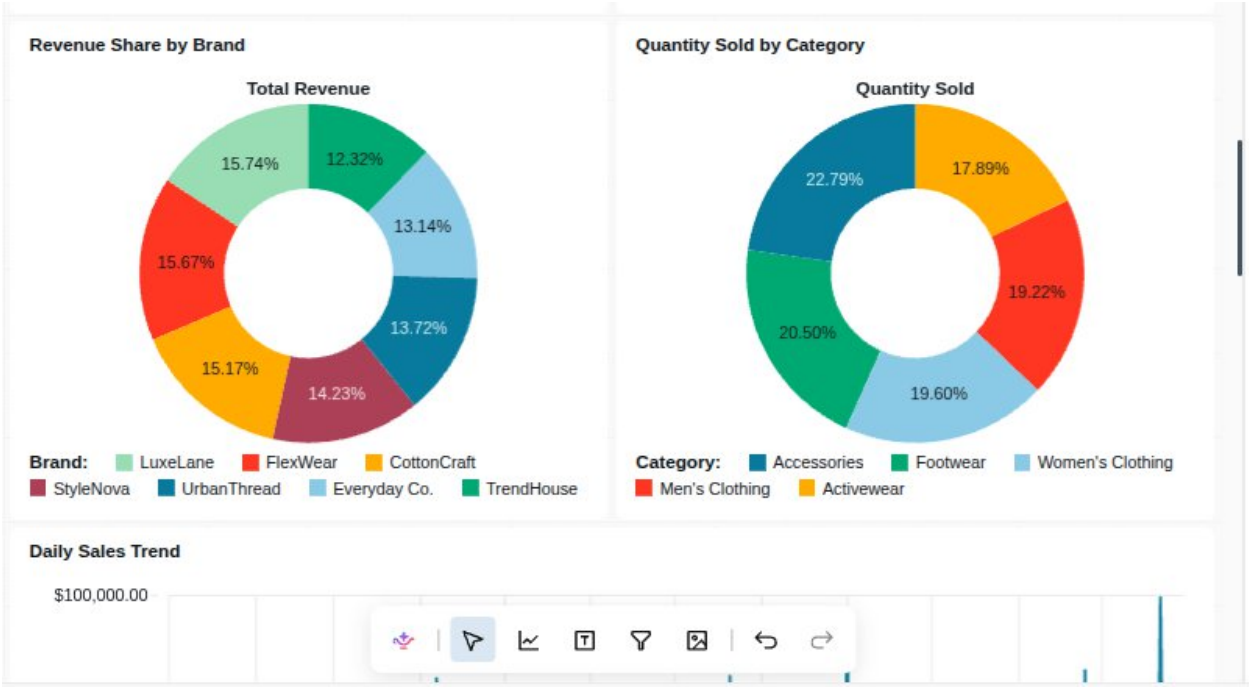
Table +

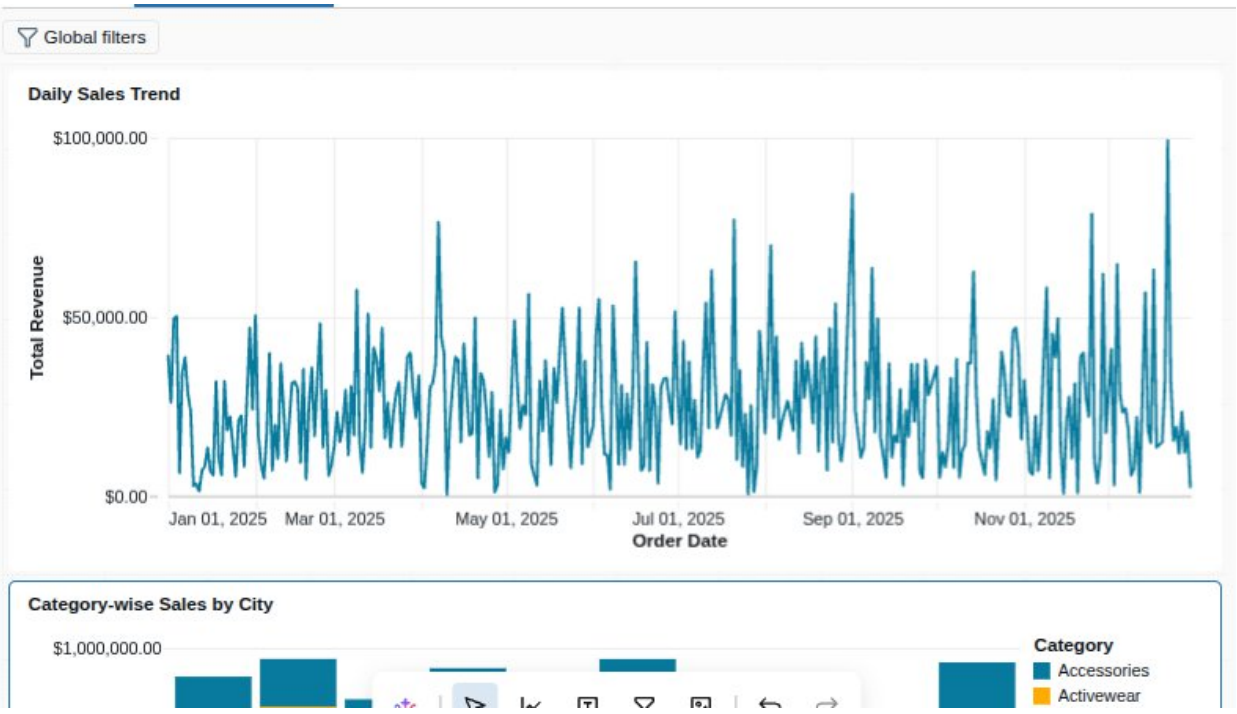
	average_selling_price
1	5252.33

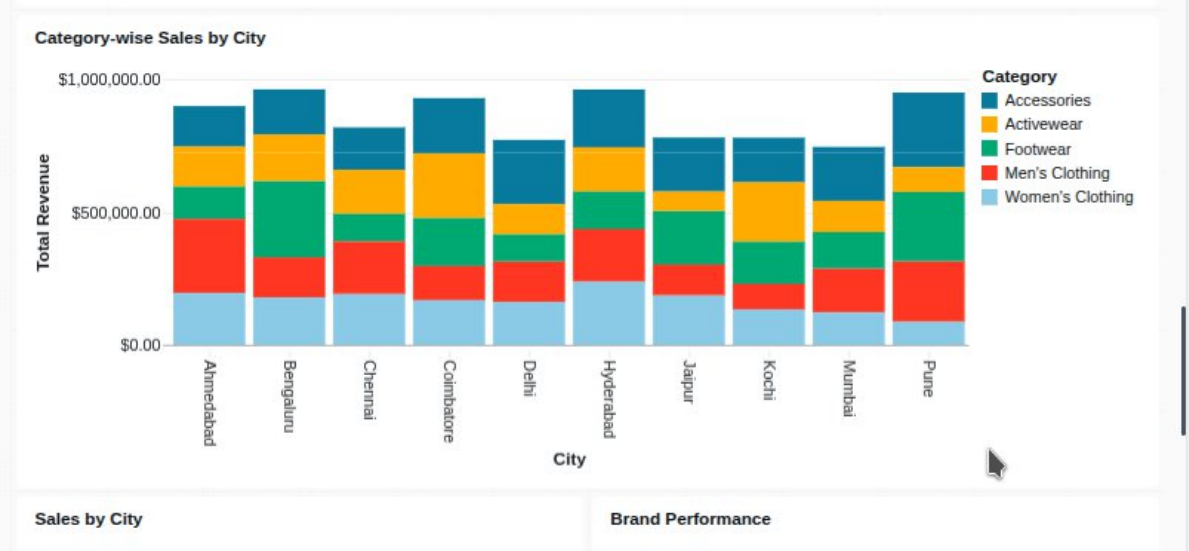
Step 4:

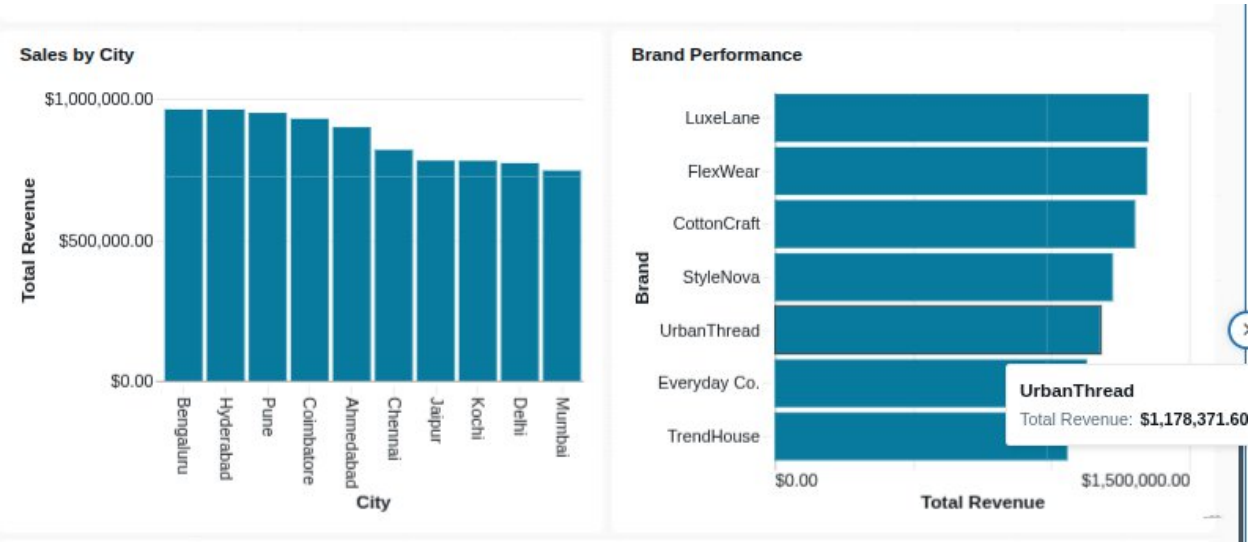

Fashion Sales Performance Dashboard 2026...
Publish
Share
Data
SalesDashboard
Data Model
Relationships
Datasets
apparel_sales Local
Add data
Add SQL dataset
UI
Overview
Measures (4)
Fields (14)
Filter
Add parameter
Join
Source
apparel_sales
Measures (4)
Count
Total Revenue
Total Quantity Sold
Average Selling Price
Fields (14)
OrderID
OrderDate
CustomerID
Preview
OrderID
OrderDate
CustomerID
CustomerName
1 ORD-202505-81482 2025-05-06 CUST-2424 Nila Nair
2 ORD-202509-95181 2025-09-16 CUST-9928 Arjun Iyer
3 ORD-202503-38221 2025-03-23 CUST-6514 Diya Sharma
4 ORD-202511-80284 2025-11-06 CUST-3045 Arjun Sharma
5 ORD-202512-20458 2025-12-27 CUST-4814 Diya Nair
6 ORD-202507-94939 2025-07-01 CUST-2169 Nila Reddy
7 ORD-202505-38785 2025-05-19 CUST-6313 Aarav Iyer
8 ORD-202504-51245 2025-04-19 CUST-4483 Kavya Nair
9 ORD-202505-86622 2025-05-07 CUST-8019 Nila Nair
10 ORD-202502-92240 2025-02-16 CUST-3621 Arjun Sharma
11 ORD-202505-99166 2025-05-09 CUST-2876 Rohan Patel
12 ORD-202503-44522 2025-03-22 CUST-9201 Ishaan Sharma
13 ORD-202503-10074 2025-03-24 CUST-6310 Kavya Kumar
14 ORD-202505-73699 2025-05-04 CUST-2133 Rohan Reddy
15 ORD-202509-80686 2025-09-28 CUST-4295 Vihaan Nair

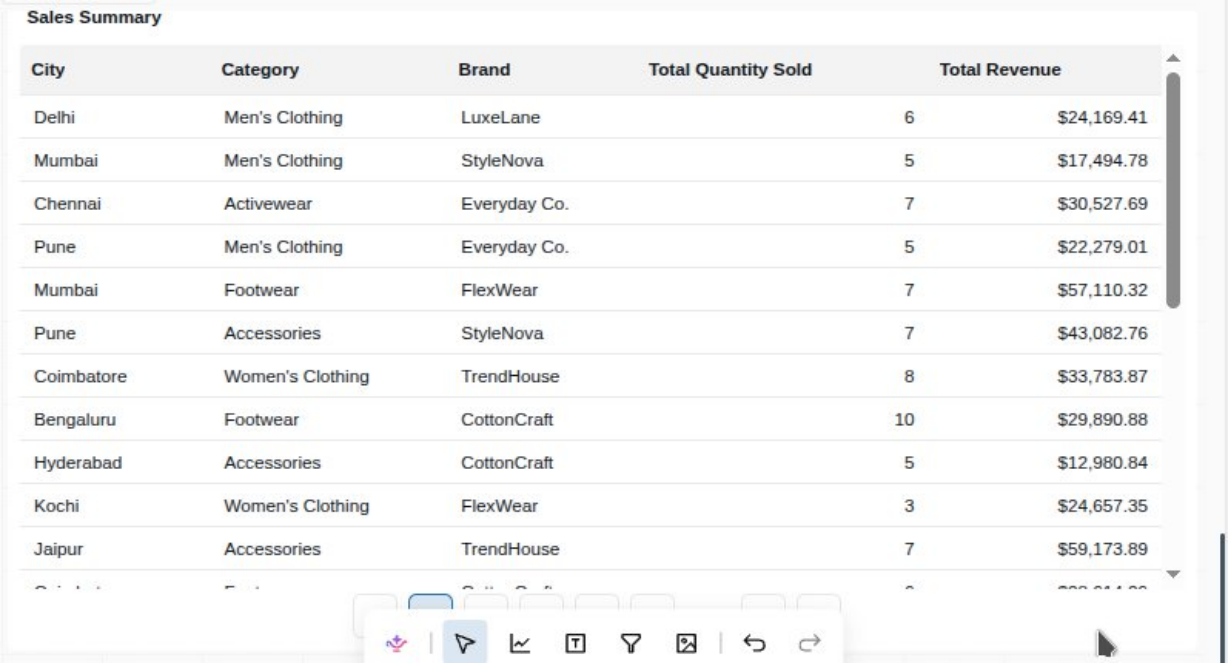

Fashion Sales Performance Dashboard 2026...
Data
SalesDashboard
Global filters
Total Sales Revenue
$8,590,530....
Total Quantity Sold
1,878
Average Selling Price
$5,252.33
Sales by Category
$2,000,000.00
$1,000,000.00
$0.00
Total Revenue
Accessories
Men's Clothin
Top 10 Products by Revenue
Product
CottonCraft Jackets
StyleNova Jackets
LuxeLane Jeans
FlexWear Sneakers
LuxeLane Formal Shoes
Everyday Co. Jackets
StyleNova Belts
UrbanThread Shorts


Revenue Share by Brand
Total Revenue
15.74%
12.32%
13.14%
15.67%
13.72%
15.17%
14.23%
Brand: LuxeLane FlexWear CottonCraft
StyleNova UrbanThread Everyday Co. TrendHouse
Quantity Sold by Category
Quantity Sold
22.79%
17.89%
19.22%
20.50%
19.60%
Category: Accessories Footwear Women's Clothing
Men's Clothing Activewear
Daily Sales Trend
$100,000.00


Global filters
Daily Sales Trend
$100,000.00
$50,000.00
$0.00
Total Revenue
Jan 01, 2025
Mar 01, 2025
May 01, 2025
Jul 01, 2025
Sep 01, 2025
Nov 01, 2025
Order Date
Category-wise Sales by City
$1,000,000.00
Category
Accessories
Activewear


Category-wise Sales by City
$1,000,000.00
$500,000.00
$0.00
Total Revenue
Ahmedabad
Bengaluru
Chennai
Coimbatore
Delhi
Hyderabad
Jaipur
Kochi
Mumbai
Pune
City
Category
Accessories
Activewear
Footwear
Men's Clothing
Women's Clothing
Sales by City
Brand Performance


Sales by City
$1,000,000.00
$500,000.00
$0.00
Total Revenue
Bengaluru
Hyderabad
Pune
Coimbatore
Ahmedabad
Chennai
Jaipur
Kochi
Delhi
Mumbai
City
Brand Performance
LuxeLane
FlexWear
CottonCraft
StyleNova
UrbanThread
Everyday Co.
TrendHouse
Brand
UrbanThread
Total Revenue: $1,178,371.60
$0.00
$1,500,000.00
Total Revenue



Sales Summary

City	Category	Brand	Total Quantity Sold	Total Revenue
Delhi	Men's Clothing	LuxeLane	6	$24,169.41
Mumbai	Men's Clothing	StyleNova	5	$17,494.78
Chennai	Activewear	Everyday Co.	7	$30,527.69
Pune	Men's Clothing	Everyday Co.	5	$22,279.01
Mumbai	Footwear	FlexWear	7	$57,110.32
Pune	Accessories	StyleNova	7	$43,082.76
Coimbatore	Women's Clothing	TrendHouse	8	$33,783.87
Bengaluru	Footwear	CottonCraft	10	$29,890.88
Hyderabad	Accessories	CottonCraft	5	$12,980.84
Kochi	Women's Clothing	FlexWear	3	$24,657.35
Jaipur	Accessories	TrendHouse	7	$59,173.89

Three Key Business Insights:

1. Top-Performing Categories: Identify which product categories generate the highest revenue to prioritize inventory and marketing efforts.

2. Best-Selling Products and Brands: Determine the top-performing products and brands by revenue to focus on popular items and improve sales strategies.

3. City-Wise Sales Performance: Compare revenue across cities to identify high-performing locations and cities with growth potential, helping businesses plan targeted promotions.